

Graph Caption Framing Student Handout

TOK Cognitive Science Activity Bank • Knowledge and Language • 20-30 min

Category	Knowledge and Language	Time	20-30 min
Difficulty	Easy	ID	graph-caption-framing

Big idea: Data needs language before it becomes a public knowledge claim.

Student-Facing Prompt

Inspect Graph Caption Framing Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Graph Caption Framing Stimulus A	Student-facing stimulus 1: Line graph: screen time rises during exam week and falls afterward. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Graph Caption Framing Stimulus B	Student-facing stimulus 2: Caption contrast: 'Students lose balance during exams' versus 'Temporary study routines shift during assessment week'. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Use this for comparison, a second round, or a partner challenge.
Graph Caption Framing Reveal / Twist	Student-facing stimulus 3: Caption contrast: 'Students lose balance during exams' versus 'Temporary study routines shift during assessment week'. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Teacher reveal: connect the changed response to the big idea — Data needs language before it becomes a public knowledge claim.

Actual Lesson Questions

#	Question for students
1	After inspecting Graph Caption Framing Stimulus A, what is your first knowledge claim?
2	Which exact detail from Graph Caption Framing Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	When does language become part of evidence?

Ready-to-Use Examples

- Line graph: screen time rises during exam week and falls afterward.
- Caption contrast: 'Students lose balance during exams' versus 'Temporary study routines shift during assessment week'.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Language.

Activity Steps

1. Show the graph without title or caption and ask students what it might show.
2. Groups write captions in assigned framing styles.
3. Students compare how each caption changes interpretation and trust.
4. Groups identify what extra context the graph needs.
5. Debrief the difference between data, interpretation, and persuasion.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: When does a caption become part of the evidence?
- Can accurate language still mislead?
- What responsibility do communicators have when turning data into claims?

Knowledge Questions

- When does language become part of evidence?
- How can accurate data be framed misleadingly?
- What makes a public knowledge claim responsible?

Transfer Example

For example, run this activity with the same graph captioned as crisis, progress, uncertainty, or manipulation. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...